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Introduction

This proposal is for a project at the Burbank Public Library (BPL). The project has two parts. The first part of the project is to build a children's STEM Lab inside the main branch of BPL. The Lab will open with two stations, with potential for growth. The second part of the project is to create one weekly program for the STEM Lab on an ongoing basis. One station in the Lab will be a collection and assortment of LEGOs that can be used by themselves or in conjunction with other STEM Lab tools. The second station will be a computer coding station, where kids will learn coding through games. The weekly program will be an after-school computer coding class taught by a local volunteer instructor.

Institution Details

The Burbank Public Library is located in Burbank, California, a small, independent city within the City of Los Angeles. This library serves the residents of Burbank and the Greater Los Angeles Area. Burbank's current population is approximately 107,000 (Burbank Public Library). According to the United States Census Bureau, 21.7% of the population is under 18. In addition to local private schools, Burbank Unified School District is the public school system located within the city and enrolls approximately 14,300 students. Despite already offering 488 YA and children's programs yearly (Institute of Museum and Library Services), during a survey in 2022, results suggested patrons were interested in more children's and teen programming, including STEM programs (Burbank Public Library). Of the nearly 10,000 attendees to BPL's programs, almost 8,000 were young adults and children (Institute of Museum and Library Services). This library already leans into providing programs for its youth patrons, who are arguably its largest resident stakeholders, and its patrons feedback confirm that they want to see more of that.

Project Topic

Maker Spaces and STEM Labs continue to be a growing trend in libraries. The composition of a maker space can vary from library to library and in different contexts, but generally speaking they are made up of different technology machines. These machines can range from 3D Printers and laser cutters, to sewing machines, Cricut machines, coding machines and more. Not only are the machines made accessible to patrons during open library hours, many spaces also provide workshops or classes for patrons to learn how to use the machines on their own (Mersand, p.175).

Maker Spaces provide many benefits to patrons. These spaces enable patrons to access tools that allow them to create, learn and produce rather than consume (Mersand, 2021, p. 174) that they might not have any other means of accessing. The space itself can be an equalizer in the digital and maker divide. Julian and Parrot state that “librarians continually search for ways to engage students in thinking, creating, sharing, and growing” and maker spaces are an obvious way for librarians to continue to do this (2017, p. 14). Maker spaces also support patrons' interest-driven learning, creative problem-solving and information/media literacy development (Kim, Jung, and Cho, 2022). This means a patron can take a class to learn to use a machine within a maker space and then come back to the library during open hours to use that machine on their own. These spaces within libraries enable patrons to become independent creators, expanding their skillsets and creative capabilities.

Maker spaces geared toward children have an even bigger impact. In addition to the above benefits, classes in maker spaces “help prepare students with valuable skills needed for college and career readiness” (Love, Roy, and Marin, 2020, p. 23) They also change student perspectives on what science and engineering jobs are and considering them as a future career path (Wall et

al, 2024, p. 1385). A STEM Lab can encapsulate all the benefits of a maker space and open young minds to endless possibilities.

Project Description & Rationale

BPL's Strategic Plans number one goal is to facilitate lifelong learning and discovery, with an emphasis on early literacy, children's development, and supporting student success. A STEM Lab geared toward children, with an introductory weekly program at BPL, is an obvious way to incorporate important aspects of a maker space with the goals of this particular library. This project very clearly supports and expands on these goals.

The benefits of this projects are many. First and foremost, this project benefits the children who participate in the program, increasing their early literacy, enabling access to critical ways of creating and engaging with STEM fundamentals. Children who use the STEM Lab are beginning a journey on path toward lifelong learning. The parents of the children will also benefit by having an additional resource for enrichment in their children's lives. These parents may not have the resources to engage their kids in a STEM space outside of the library. The community as a whole benefits with increased literacy rates and a higher level of educated kids growing up in the community. In addition, the library benefits from having more ways to connect with patrons, more services offered, and more patrons using the space.

Project Goals, Objectives, & Activities

The goal of this project is to support the library's strategic plans by creating an interactive learning experience to elevate accessibility for children's early literacy. The objective is to create a STEM Lab with two stations and one program class a week. The Lab stations will be a LEGO station and a computer coding station. LEGOs are a fun activity for children by themselves and a fundamental in STEM skills (Blackburn, 2016). BPL already hosts a weekly LEGO program for

young children, but it is not a permanent station within the library. Creating a permanent Lab station for the LEGOS allows kids and families to have more regular access to them and also saves staff time and energy from not having to drag them out and put away for each program.

The second station in the Lab will be a computer coding station. The library would purchase or reroute existing computers to be set up in the Lab for the purpose of computer coding. Many Labs use computer games to teach children coding and programming, and it is also common for Labs to use LEGOS and coding to create robots (Mersand, 2021, p. 175).

Starting the Lab with these two stations uses resources the library already has efficiently and also doubles the use of each station, which can be used independently or together.

The STEM Lab itself will need to be established in a space within the library. This will require collaborating with the library management on what area can be used and potentially clearing out what is already existing in that space. It will also require research and purchasing of Lab equipment and supplies. To establish a weekly class, the project manager will need to find collaborators in the community. This would require researching and connecting with teachers or subject experts who would like to participate in this program and teach a weekly class.

Once a space is found and collaborators are confirmed, the space needs to be built. This could be library staff building new tables or furniture for the Lab equipment to be stationed or hiring outside labor to install them. It may also require help from experts, perhaps the teacher collaborators to put together the STEM stations. LEGOs may not be so difficult to set up, but help may be required setting up and install computer coding games at the new station.

Project Assessment, Evaluation, & Deliverables Communication plan

Assessment will be both quantitative and qualitative. Instructors will be asked to informally ask the kids in the program at the end of the class if they learned anything. Based on

students' responses, they will make notes to be kept for later evaluation. Parents of students in the class will be asked to take a brief paper or digital survey with less than ten questions. The survey evaluators are aware that parent feedback is not necessarily a formal and substantive evaluation of their child's progress, but is a good benchmark for the library to use for assessment given the lack of formal structure for this space. Paper surveys will be available in the Lab and a QR code provided for those who want to take it digitally. The survey will ask if they think their child has learned new skills during their time in the program, as well as suggestions for improvement to the Lab or the program they attended. The instructor will also be asked to take attendance during each class and attendance numbers will be recorded. A notebook or spreadsheet will be set up for instructors to easily take these notes for later evaluation.

The results of these three assessment methods will be evaluated every four months. The program manager will evaluate the qualitative data for any feedback that might suggest adjustments or improvements to the lab or the program. They will also look at attendance and confirm if the Lab programs are popular enough to warrant the program continuing and possibly adding more weekly programs. Feedback may also indicate what additional tools or stations patrons may be interested in exploring and what new types of weekly programs should be considered.

Evaluating the Lab every four months allows the manager to not be overwhelmed with having to constantly review assessment material and also not having to invent a new station or program too often. However, it is often enough that if there is a need for improvement, it will be addressed in a timely manner.

There are several deliverables for this project. The first deliverable is a budget and proposal for the required equipment and installation of the lab. This would include pricing for

equipment needed and a person hour estimate of how long the set up will take. Another deliverable is the assessment, which includes preparing the questions, creating the physical assessment and QR link. A notebook or spreadsheet to track assessment will be required for this deliverable. Along with this, an additional deliverable would be a summary of the evaluation of the assessment and written guidelines for adjustments or additions to the Labs ongoing use, which will be required on an ongoing four-month basis.

The communication plan also includes a deliverable which is a marketing plan to market the new Lab. This plan will ensure that patrons are aware that this new space available for use and when the weekly programs will take place. One aspect of the marketing plan deliverable would be creating a flyer to be delivered to local schools. This flyer will go directly to the stakeholders (children), making them aware of the new program, and bringing that information to their parents to encourage their participation. Another aspect of this communication plan is to make sure the rest of the library staff knows about the Lab and the program. The project manager will likely have to work out the program schedule with the library programming manager.

Project Resources

The resources for this project will come from various sources. The first will be financial support to purchase the needed equipment. As previously stated, efforts will be made to utilize equipment that the library already has, but there will be additional expenses as well. This financial support may come from a grant or from the library budget, which may be able to allocate part of its budget to start this Lab. Purchases will include furniture for stations, additional LEGO sets, and computer coding programs.

Another resource needed is for the labor of staff to set it up the physical space. This may be library employees or could be outsourced labor depending on how complicated the installation

winds up being. Library staff will also need to be trained to learn how to use and instruct with the Lab equipment. The training itself may need financial resources as well. The program teacher is intended to be a volunteer but may need a stipend for ongoing services or training.

Project Timeline

This Lab will take approximately six months to set up and then will be open on an ongoing basis, with the aforementioned four-month evaluations. The process will begin with initial research of the best equipment, tools and needs of a STEM Lab and creating the budget and program plan for the Lab. This will take approximately two months. Next, the library will begin ordering equipment, reaching out to STEM teachers and instructors at local schools for help and collaboration, creating the marketing plan and marketing materials, and installation plan, which will also take approximately two months. The final two months will be the installation, training of staff, and implementation of the marketing plan. At the end of the six months, the STEM Lab will be up and running and updates or additions to the program classes are what will be required to continue its success.

Conclusion

The Burbank Public Library is a well-loved and used library with room to grow in its services for youth. The proposed STEM Lab will be a welcome addition to the library and fulfills both the libraries plan to invest in early childhood literacy and the interests of the community. The benefits of this Lab are many, including access and enrichment for the children who use the library and the greater community, as well as higher patron use for the library. Overall, this project is a welcome addition to the Burbank Public Library.

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